

# FREDERYCO MIGUEL MUNIZ BELÉM

Senior Backend Engineer · PHP / Laravel · Node.js / TypeScript · APIs & Integrations · AI: Agents & MCP

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## PROFESSIONAL SUMMARY

Senior Backend Engineer with 13+ years designing and evolving critical production systems, with strong expertise in API architecture (REST and SOAP), distributed integrations, and high-complexity relational databases (PostgreSQL, Oracle, MySQL, SQL Server). Hands-on with Node.js, AWS (S3, Lambda, EC2), Terraform, Docker, Kubernetes fundamentals, and asynchronous processing (queues) in high-demand environments. Autonomous in architecture decisions, applying SOLID and design patterns, legacy modernization, and microservices migration, with measurable performance and availability results. Advanced use of AI in the engineering workflow — code agents, prompt/context engineering, orchestration of multiple specialized agents (sub-agents), and MCP (Model Context Protocol) servers in production. Experience in technical leadership, mentoring, and agile practices (Scrum).

## KEY RESULTS

- Oracle integration with index and query optimization — **60% reduction in query time** on the Hemominas SGPR system
- Hemominas SGPR (Laravel/Oracle) with scalable architecture for **10M+ records per year** and hospital-system integration
- Architected an **MCP (Model Context Protocol)** server and a sync tool (Node.js + Claude API) that eliminated manual rework between Azure DevOps and GitLab/SharePoint, with a 3-layer fallback
- **+40%** API performance and **-40%** deployment time through Docker, CI/CD, and automated testing
- Magento e-commerce supporting seasonal peaks of **1,000+ requests/second**

## PROFESSIONAL EXPERIENCE

### Senior Software Developer — Montreal Official · Remote

May 2024 — Sep 2026

- Backend services and **REST APIs in PHP (Laravel)** and **PostgreSQL** for client FonteCred, with authentication/authorization and access control, in a fully remote environment
- **IPSEMG/SIGAS project (healthcare)** with Laravel, MySQL, and Next.js/React.js: critical management platform for surgeries and critical data, integrating front-end and backend
- **MCP server** and **orchestration of multiple specialized agents (sub-agents)** integrating Azure DevOps and GitLab; automation with the **Claude API** for data extraction and elimination of repetitive manual work
- Complex integrations with **CRM, WhatsApp Business API**, and LLMs (**Gemini**, in addition to Claude) for support-flow automation and data extraction
- **SOAP Web Services** for integration with legacy healthcare systems
- Production incident analysis and resolution (troubleshooting and root-cause) on critical systems
- **Code review**, developer mentoring, and definition of backend standards (SOLID, design patterns)
- **Docker, CI/CD** (GitLab/Bitbucket), and automated testing — 40% faster deployments  
**Stack:** PHP (OOP/Laravel), PostgreSQL, Node.js, REST/SOAP, Swagger, Docker, GitLab CI/CD, Claude Code, MCP, React.js, WhatsApp Business API, Gemini

### Senior Software Engineer — Stefanini Brasil · Remote

Sep 2022 — Dec 2023

- **SGPR** module for **Fundação Hemominas (healthcare)** in **PHP/Laravel** with **Oracle/MySQL** integration and scalable architecture for 10M+ records
- **REST and SOAP APIs** documented with **Swagger/OpenAPI** for hospital-system integration
- **Oracle index and query optimization** — 60% reduction in query time; high-complexity data modeling
- Applied **SOLID, design patterns**, and engineering best practices on a large codebase
- Docker integrated with CI/CD (GitLab), automated testing, and Vue.js front-end  
**Stack:** PHP (OOP/Laravel), Oracle, MySQL, REST/SOAP, Swagger/OpenAPI, Docker, GitLab CI/CD, Vue.js

## Development Analyst III — JN2 E-Commerce Expert · Brazil

Jul 2021 — Aug 2022

- Developed and maintained **Magento** stores (mid/large clients) in **PHP (OOP)** and **MySQL**, focused on performance and scalability
  - Custom features in OOP PHP; support for seasonal campaigns with peaks of 1,000+ req/s
  - **AWS (S3), Terraform, Docker, Bitbucket CI/CD** ensuring consistent deployments
- Stack:** PHP (OOP/Magento), MySQL, AWS, Terraform, Docker, Bitbucket CI/CD

## Web Developer (incl. Tech Lead) — CRP Tecnologia · Palmas, TO

Feb 2018 — Jul 2021

- Web applications in **PHP (OOP: Laravel)** and **C# (.NET/Umbraco)** for public and private clients
  - **Tech Lead (Jul–Dec 2020):** main point of contact for the outsourcing team, Scrum ceremonies, and structured feedback
  - **REST and SOAP Web Services** with **CRM** integration and relational databases (SQL Server, MongoDB); Swagger/OpenAPI documentation and automated testing
- Stack:** PHP (OOP/Laravel), C#/.NET, SQL Server, MongoDB, Docker, Azure, Cloudflare

## Web Developer — ArtemSite Agência Digital · Palmas, TO

Sep 2015 — Jan 2018

- Backend development of websites and portals in **PHP (OOP)** and **MySQL**; system integration and internal workflow automation
- Stack:** PHP (OOP), MySQL, Docker, jQuery

## Internships — DIX BPO (2014–2015) and CREA-TO (2012–2014) · Palmas, TO

- Institutional websites and portals in **PHP + MySQL + Bootstrap**; TOTVS integration via APIs/ADVPL; first projects using **Git**

## TECHNICAL SKILLS

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**Languages:** PHP (OOP), Node.js, JavaScript/TypeScript, Java, C#, SQL, Bash, Python (developing)

**Frameworks:** Laravel, Symfony, Spring Boot, .NET, Next.js, React.js, Vue.js, Magento

**Engineering & Patterns:** SOLID, design patterns (GoF), Clean Code, TDD, code review, legacy refactoring and modernization

**APIs & Security:** REST, SOAP, XML, OpenAPI/Swagger, Microservices, authentication/authorization (OAuth2/JWT), RBAC, API security

**Databases:** PostgreSQL, Oracle, MySQL, SQL Server, MongoDB — modeling, queries, index optimization

**AI & Agents:** Claude Code (code agents), prompt/context engineering, orchestration of multiple specialized agents (sub-agents), MCP (Model Context Protocol), Claude API, Gemini, LLM-based workflow automation

**DevOps & Cloud:** Docker, CI/CD (GitLab, Bitbucket), AWS (S3, Lambda, EC2), Terraform, Kubernetes (fundamentals), Azure, Cloudflare

**Architecture:** architecture decisions, microservices, asynchronous processing / queues, event-driven integrations

**Agile & Tools:** Scrum, technical mentoring, Git/GitHub, PhpStorm, Postman, Linux, Nginx, Apache

## EDUCATION

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- **Postgraduate** — Specialization in Software Development with DevOps Practices — Cruzeiro do Sul Virtual · Aug 2023 – Sep 2024
- **Bachelor's Degree** — Information Technology / Information Systems — UniCatólica · Jan 2012 – Jun 2016

## CERTIFICATIONS & COURSES

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- Claude Academy: Claude Code 101 — Anthropic, 2026
- AI-Driven Development Workshop — Tech Leads Club, 2026
- Tech Lead Career: Technical Leadership — Alura, 2024
- DevOps & Software Development Practices — Cruzeiro do Sul Virtual, 2024
- Docker & Kubernetes Fundamentals
- Scrum Foundation Professional Certificate (SFPC) — Certiprof, 2020
- AWS Cloud Practitioner (in progress)

- Laravel Advanced Patterns & Best Practices

## LANGUAGES

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- **Portuguese** — Native
- **English** — Technical (reading and writing)